Alessandro **Santini**

PhD candidate in Gravitational Physics

Max Planck Institute for Gravitational Physics, Am Mühlenberg 1, D-14476 Potsdam, Germany

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Education

Ph.D. in Physics

Potsdam, Germany

Max Planck Institute for Gravitational Physics — *2023–present*

- **Thesis:** The LISA Global Fit
- **Supervisor:** Dr. Jonathan Gair

Master's Degree in Astrophysics and Space Physics

Milan (MI), Italy

Università degli studi di Milano-Bicocca — *2021–2023*

- **final degree grade:** 110/110 with distinction (cum laude)
- **Thesis:** Black-hole mergers in disk-like environments could explain the observed $q - \chi_{\rm eff}$ correlation
- **Supervisors:** Prof. Davide Gerosa, Dr. Roberto Cotesta

Bachelor's Degree in Physics

Milan (MI), Italy

Università degli studi di Milano-Bicocca — *2018–2021*

- **final degree grade:** 110/110 with distinction (cum laude)
- **Thesis:** Resolution of the Euler equations using the Athena++ code
- **Supervisor:** Prof. Bruno Giacomazzo

High School Diploma

Lissone (MB), Italy

Liceo Scientifico Statale Federigo Enriques — *2013–2018*

- **Final degree grade:** 97/100

Internships

Research visit

Zurich, Switzerland

ETH Zurich, 1.5 months — *November 2026*

Research visit

Nice, France

Observatoire de la Côte d'Azur, 1 month — *May 2026*

Erasmus+ Scholarship

Baltimore (MD), USA

Johns Hopkins University, 3 months — *April–July 2023*

Extracurricular courses and workshops

Data visualization workshop

Potsdam, Germany

Visualize your science – https://www.visualizeyourscience.com/ — *September 2026*

Skills

Programming Python (proficient) – GPU computing (CUDA, Torch, Cupy, Numba, JAX) – Bash – Mathematica (basic) – C, C++ (basic)

Other tools Latex (proficient), Git, Microsoft Office suite

Languages Italian (Native), English (Fluent)

Talks, conferences & workshops

Invited Talks

De Tenebris ERC kick-off meeting — *GRAPPA, Amsterdam, The Netherlands*

Global fit, EMRIs, and other nightmares — *November 2026*

LISA AI Hackathon — *ESTEC, Noordwijk, The Netherlands*

Machine Learning in the Global Fit – DISCUSSION SESSION — *July 2026*

Observatoire de la Côte d'Azur — *OCA, Nice, France*

Advancing LISA Global fit prototypes — *May 2026*

ETH Zurich – Gravitational Physics Group — *ETH, Zurich, Switzerland*

Advancing LISA Global fit prototypes — *March 2026*

LISA without frontiers — *Sexten, Italy*

Modelling noise and characterising stochastic signals in LISA – PANEL SESSION — *January 2026*

Contributed Talks

EMRI Search and Inference within the LISA Global Fit - Part II — *NUS, Singapore*

Multi-source analysis of EMRIs in the Mojito light common dataset — *July 2026*

26th SIGRAV Meeting — *Unimib, Milan, Italy*

A flexible, GPU-accelerated approach for the joint characterization of LISA instrumental noise and Stochastic Gravitational Wave Backgrounds — *September 2025*

EMRI Search and Inference within the LISA Global Fit - Part I — *APC, Paris, France*

Don't reinvent the wheel: including Extreme Mass Ratio Inspirals in the LISA global fit — *June 2025*

Amaldi15 — *Online*

Black-hole mergers in disk-like environments could explain the observed $q - \chi_{\rm eff}$ correlation — *July 2023*

APS April Meeting — *Minneapolis, USA*

Migration traps in AGN disks and hierarchical mergers as promising origin of the observed $q - \chi_{\rm eff}$ correlation — *April 2023*

Contributed Posters

16th LISA Symposium — *Baltimore (MD), USA*

Advancing LISA Global Fit prototypes in the time-frequency domain — *June 2026*

15th LISA Symposium — *UCD, Dublin, Ireland*

A flexible approach for the joint characterization of LISA instrumental noise and Stochastic Gravitational-Wave Backgrounds — *July 2024*

Publication record

1. **Santini, A.**; Bandopadhyay, D.; Gair, J.
   "Phentax: A JAX implementation of the IMRPhenomTHM waveform model", ArXiv preprint
2. Speri, L.; Duque, F.; Barsanti, S.; **Santini, A.**; Kejriwal, S.; Burke, O.; Chapman-Bird, C. E. A.
   "Quantifying the Scientific Potential of Intermediate and Extreme Mass Ratio Inspirals with the Laser Interferometer Space Antenna", Phys.Rev.D 114, 064012
3. De Santi, F.; **Santini, A.**; Toubiana, A.; Karnesis, N.; Gerosa, D.
   "Inferring the population properties of galactic binaries from LISA's stochastic foreground", Phys.Rev.D 113, 123006
4. **Santini, A.**; Muratore, M.; Gair, J.; Hartwig, O.
   "A flexible, GPU-accelerated approach for the joint characterization of LISA instrumental noise and Stochastic Gravitational Wave Backgrounds", Phys.Rev.D 112, 084050
5. Chapman-Bird, C. E. A. et al (including **Santini, A.**)
   "The Fast and the Frame-Dragging: Efficient waveforms for asymmetric-mass eccentric equatorial inspirals into rapidly-spinning black holes", Phys.Rev.D 112, 104023
6. Fabbri, C. M.; Gerosa, D.; **Santini, A.**; Mould, M.; Toubiana, A.; Gair, J.
   "Reconstructing parametric gravitational-wave population fits from non-parametric results without refitting the data", Phys.Rev.D 111, 104053
7. Khalvati, H.; **Santini, A.**; Duque, F.; Speri, L.; Gair, J.; Yang, H.; Brito, R.
   "Impact of relativistic waveforms in LISA's science objectives with extreme-mass-ratio inspirals", Phys.Rev.D 111, 082010
8. **Santini, A.**; Gerosa, D.; Cotesta, R.; Berti, E.
   "Black-hole mergers in disk-like environments could explain the observed $q - \chi_{\rm eff}$ correlation", Phys.Rev.D 108, 083033

Public Outreach

Long Night of Astronomy — *Berlin, Germany*

Sounds of silence: 10 Years of gravitational-wave astronomy. — *September 2025* — *Language: English*

Speaker at the closing event of the Long Night of Astronomy 2025 (webpage here), presenting the last 10 years of gravitational-wave astronomy together with 8 other researchers from the Max Planck Institute for Gravitational Physics. Audience: > 100 people.

Berlin Science Week

Berlin, Germany
November 2024
Language: English

Deciphering black hole symphonies: The new world of gravitational wave astronomy.

Speaker at one of the highlight, invited events of Berlin Science Week 2024 (webpage here), presenting the current status of gravitational-wave astronomy together with 7 other researchers from the Max Planck Institute for Gravitational Physics. Audience: > 200 people.